NEXONIX REFLEXO PVT. LTD.

CAD Designing Internship 2026

INTERNSHIP REPORT

ON

“CAD DESIGNING — SKETCHING,

PART MODELLING & ASSEMBLY DESIGN”

Submitted in partial fulfilment of the requirements of the

CAD Designing Internship Program

Submitted by

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Certificate

Student Declaration

I, Suraj Singh, bearing Roll Number [2407340400051], a student of B.Tech (Mechanical Engineering) at Rajkiya Engineering College, Banda, affiliated to Dr. A.P.J. Abdul Kalam Technical University, Lucknow, hereby declare that the internship report titled "CAD Designing — Sketching, Part Modelling & Assembly Design" is an original record of the work carried out by me during my internship at Nexonix Reflexo Pvt. Ltd.

The work presented in this report was completed during the internship period under the guidance of my company mentor Anmol verma , and my faculty mentor Mr Saurabh Tripathi Sir . All the models, screenshots and observations included here were produced by me. Wherever the ideas, text or data of other authors have been used, they have been acknowledged through proper references.

I further declare that this report has not been submitted, in part or in full, to any other institute or university for the award of any degree, diploma or certificate.

Place: Banda

Signature: _____

Suraj Singh

Acknowledgement

I would like to express my sincere gratitude to Nexonix Reflexo Pvt. Ltd. for giving me the opportunity to undergo this internship and gain practical exposure to computer-aided design. The experience has helped me connect the concepts learned in the classroom with real design practice.

I am deeply thankful to my company mentor, for the clear guidance, timely feedback and patient explanations throughout the four weeks. Every weekly assignment became easier because of the structured direction and encouragement received.

I extend my heartfelt thanks to my faculty mentor, for supporting my participation in this internship and for the academic foundation that made it possible for me to follow the training with confidence.

I also thank my fellow interns and teammates for the discussions, the sharing of modelling tips and the friendly environment that made learning enjoyable. Finally, I am grateful to my family and friends for their constant motivation.

Suraj Singh

Abstract

This report presents the work carried out during a four-week internship at Nexonix Reflexo Pvt. Ltd., focused on three-dimensional computer-aided design using SOLIDWORKS Premium 2021. The main objective of the internship was to build a strong foundation in parametric modelling, beginning with simple two-dimensional sketches and progressing step by step to solid parts, multi-view part design and, finally, a complete assembly.

In Week 1, the fundamentals of the SOLIDWORKS environment were learned, including the FeatureManager design tree, reference planes, sketch tools, Smart Dimension and geometric relations. Four exercises were completed: a 60 mm × 40 mm rectangular sketch, a concentric circle profile of Ø50 mm and Ø20 mm, an L-shaped profile with an overall length of 80 mm, and a solid L-bracket created from that profile with mounting holes using extrude boss and extrude cut features.

In Week 2, the modelling complexity was increased. A bent bracket with multiple holes and tabs was built with extrude, cut, reference plane and Flex features and inspected in Front, Left, Top and Isometric views. A triangular base plate with a pyramid-shaped body, slots and rounded edges was also created using extrude, shell and fillet features.

In Week 3, the parts were brought together in the assembly environment. A wheel-and-bracket assembly consisting of a base plate, support brackets, a wheel, a bush and a pin was assembled using mates, fixed and floating components, and colour appearances to distinguish the parts. Week 4 was used to review the models, resolve open issues and prepare this report.

The results were verified through sketch status indicators (Fully Defined or Under Defined), error-free feature trees, multi-view inspection and assembly mate checks. The internship improved my skills in sketching, feature-based modelling, design intent, assembly practice and technical documentation, and gave me a clear idea of how CAD is used in an industrial workflow. Future work includes fully defining all sketches, assigning materials, preparing engineering drawings and performing motion and strength studies on the assembly.

Keywords: SOLIDWORKS, parametric modelling, sketching, extrude, assembly, mates,

CAD.

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Week 1: SOLIDWORKS Basics, Sketching and First Solid Model

Objectives

The internship at Nexonix Reflexo Pvt. Ltd. was designed to give hands-on exposure to 3D CAD modelling with SOLIDWORKS. The objective of Week 1 was to gain a working command of the basic modelling workflow: opening a sketch on a reference plane, drawing entities, applying dimensions and relations, and converting a closed sketch into a solid feature. These technical skills are the base for all later work, and they turned classroom theory such as engineering geometry and design intent into practical models.

Learning Topics

- Introduction to the SOLIDWORKS interface: Features and Sketch tabs, FeatureManager design tree, Front / Top / Right planes and the Origin.
- Sketch entities: line, rectangle, circle and the Trim, Offset, Mirror and Convert Entities tools.
- Smart Dimension and geometric relations (horizontal, vertical, concentric, midpoint), and sketch status (Under Defined, Fully Defined).
- Converting sketches to solids with Extruded Boss/Base and Extruded Cut.

Assignments and Screenshots

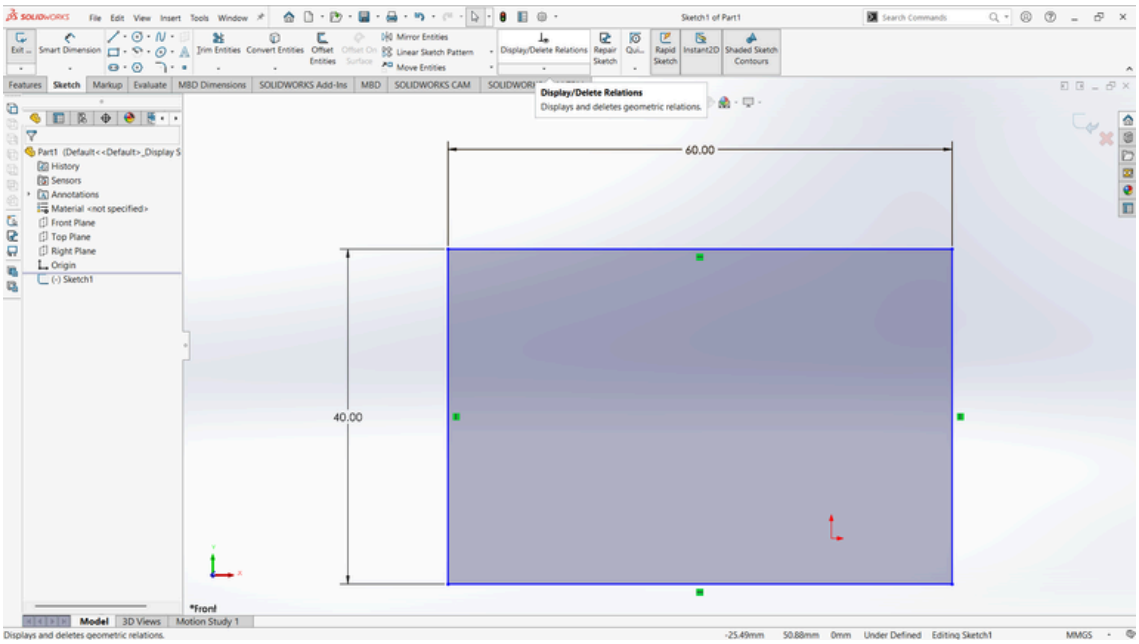


Figure 1: Rectangular sketch (60 mm × 40 mm) in Sketch1 of Part1

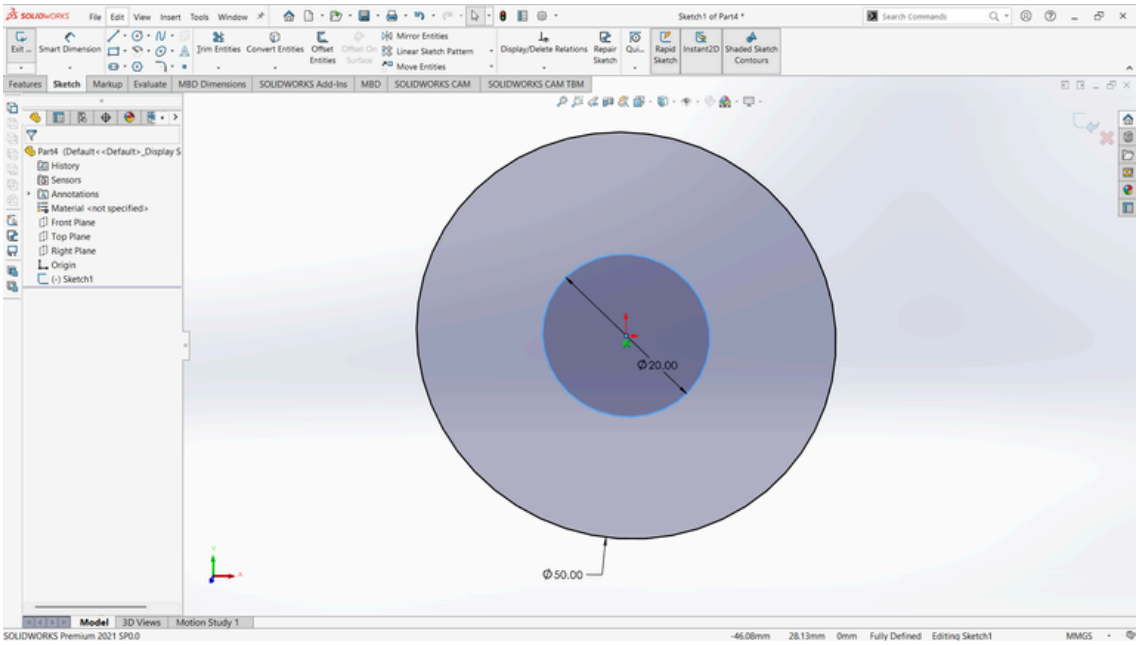


Figure 2: Concentric circle profile (Ø50 mm outer and Ø20 mm inner) in Part4

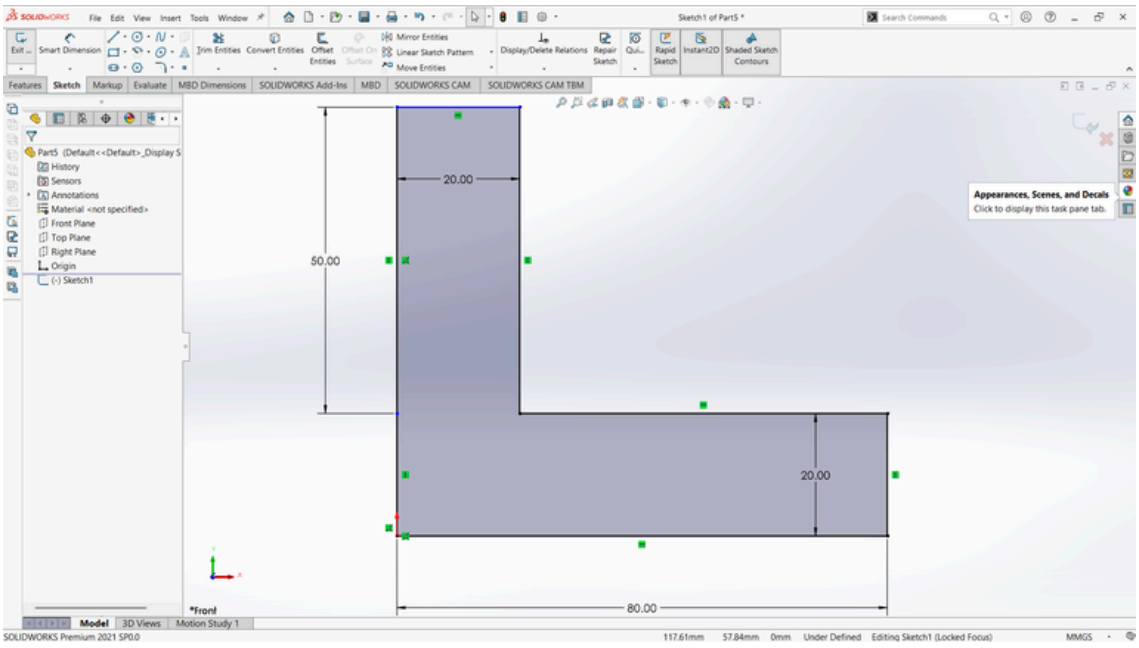


Figure 3: L-shaped profile sketch with dimensions (80, 50 and 20 mm) in Part5

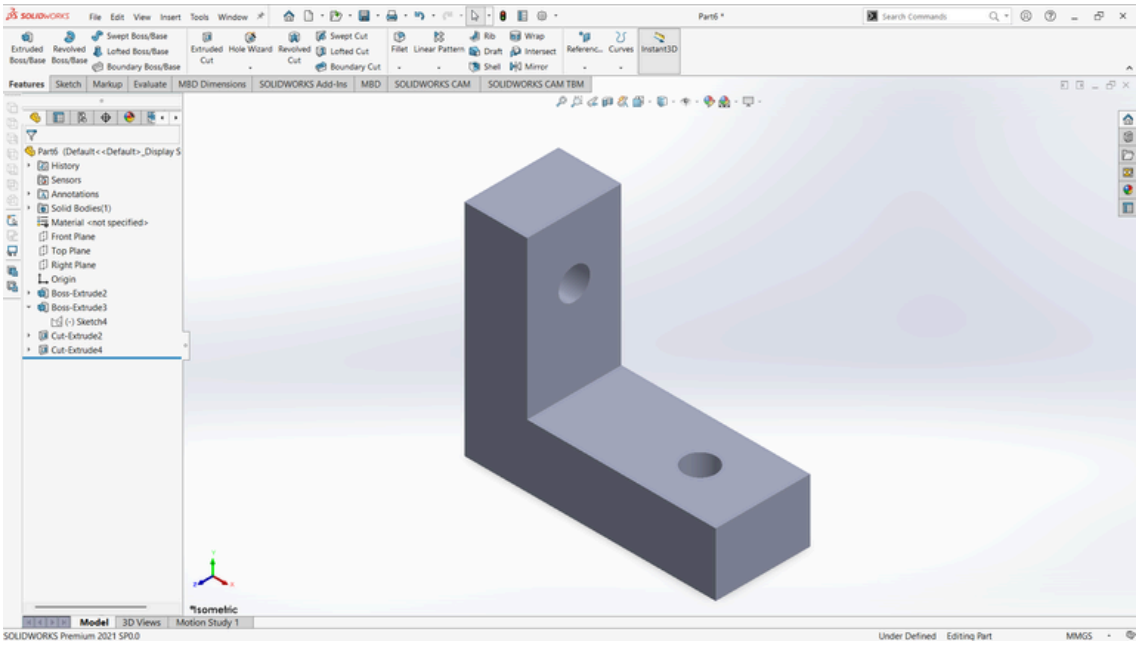


Figure 4: Solid L-bracket with two holes (Part6) in Isometric view

Testing and Results

Each model was checked using the sketch status in the status bar and the rebuild status of the feature tree. The rectangle (Figure 1) and the L-profile (Figure 3) were Under Defined, the circle profile (Figure 2) was Fully Defined, and the L-bracket (Figure 4) rebuilt as a single solid body without errors.

Challenges

The main problem in Week 1 was sketches that stayed Under Defined. Missing dimensions or relations, such as tying the rectangle and the L-profile to the Origin, had to be identified and added using Smart Dimension and the Display/Delete Relations tool. The material was also left as "not specified", so mass information could not be obtained.

Skills Learned

- Creating, dimensioning and constraining 2D sketches on a reference plane.
- Reading sketch status and understanding the difference between Under Defined and Fully Defined.
- Using Extruded Boss/Base and Extruded Cut to create solids and holes.

Outcomes

By the end of Week 1, the basic part-modelling workflow (sketch, dimension, extrude, cut) was understood, and a simple bracket with holes was modelled successfully.

Week 2: Advanced Part Modelling and Multi-View Design

Objectives

The objective of Week 2 was to move from simple sketches to parts with many features, and to check a design from every direction. This turned classroom topics such as orthographic projection into practice: the same 3D model produces the Front, Left and Top views of a drawing, and inspecting all of them together confirms that holes and edges are positioned correctly.

Learning Topics

- Working in the four-view layout (Front, Left, Top, Isometric / Trimetric) to check a part from all sides.
- Reference geometry: creating additional planes (Plane3, Plane5) for sketches at new locations.
- Flex feature for producing a bend, together with multiple extrude boss and extrude cut features.
- Shell and Fillet features for hollowing a body and rounding edges.

Assignments and Screenshots

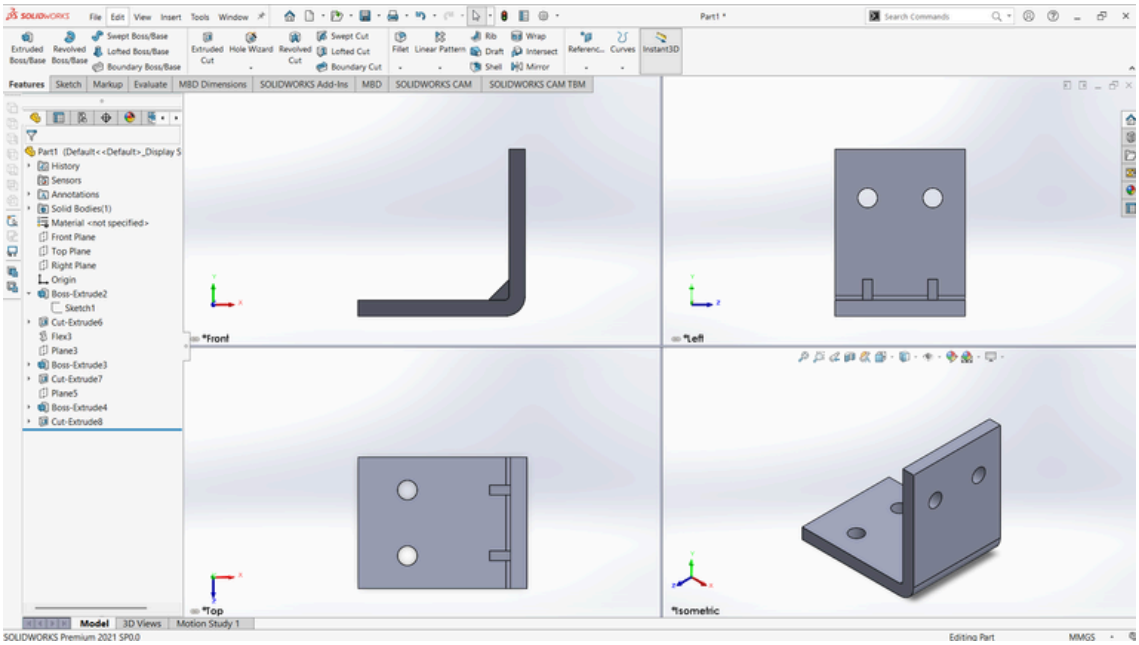


Figure 5: Bent bracket with holes and tabs (Part1) shown in Front, Left, Top and Isometric views

Figure 5 shows Part1, a bracket bent into an L-shape, displayed in four views at the same time. The Front view shows the bent profile with a rounded outer bend and a small triangular reinforcement at the inner corner. The Left view shows two circular holes in the upright flange, and the Top view shows two holes in the base flange along with small rectangular tabs on one edge. The feature tree shows how it was built: a base Boss-Extrude2 from Sketch1, several cuts (Cut-Extrude6, 7 and 8), the Flex3 feature that creates the bend, two reference planes (Plane3 and Plane5) and additional bosses (Boss-Extrude3 and 4). The four-view layout was used to confirm that the holes and tabs were positioned correctly in every direction.

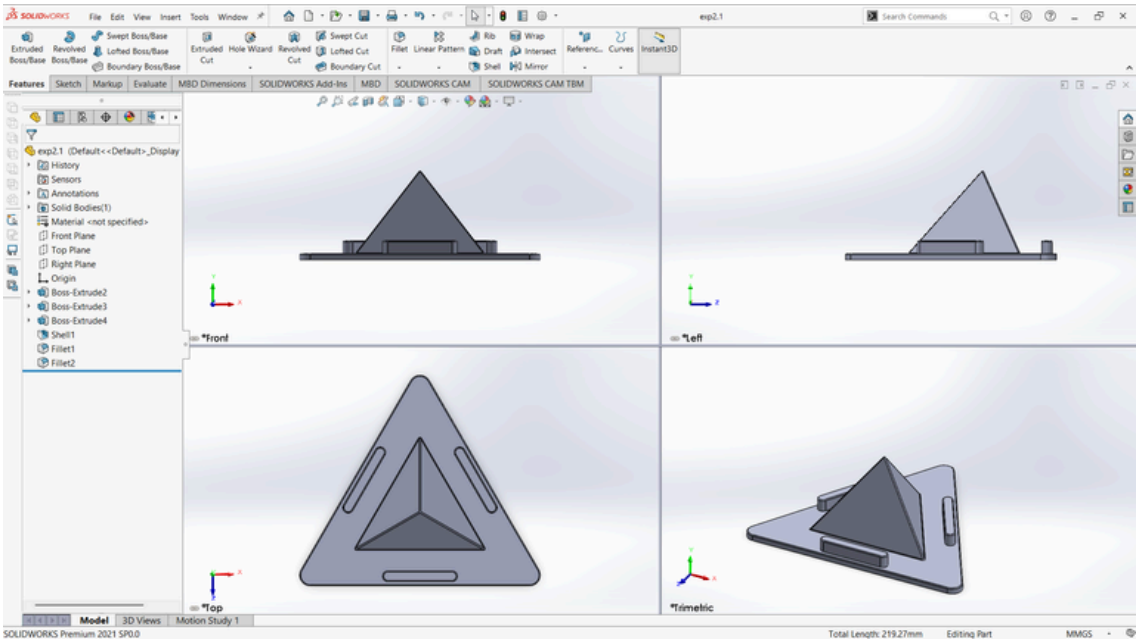


Figure 6: Triangular base plate with pyramid-shaped body and slots (exp2.1) in four views

Figure 6 shows the part exp2.1 in the Front, Left, Top and Trimetric views. It has a thin triangular base plate with rounded corners, a raised pyramid-shaped body at the centre, a long slot near the bottom edge of the plate and two slanted slots along the sides, together with small raised pads near the base of the pyramid. The feature tree contains Boss-Extrude2, 3 and 4 for the plate and raised features, Shell1 to hollow the body, and Fillet1 and Fillet2 to round the edges. The status bar shows "Total Length: 219.27mm", which is a measurement of the selected edges. This exercise combined sketching, extrusion, shelling and filleting in one part and showed how the order of features affects the final shape.

Testing and Results

Both parts rebuilt without errors in their feature trees. The bent bracket (Figure 5) was verified for holes, tabs and bend in four views, and the triangular base (Figure 6) was verified for slots, shell and fillets in four views.

Challenges

The order of features such as Flex, Shell and Fillet changed the result and had to

be planned carefully. Features also had to be placed on additional reference planes rather than on the original planes.

Team Collaboration

The internship followed a mentor-guided structure. Assignments were discussed with the company mentor, doubts were raised in review sessions, and modelling approaches were compared with fellow interns. Comparing feature trees showed that the same shape can be modelled in more than one way, and that a clean, well-named feature tree is important when someone else must open and edit the file. [Add one or two specific examples of discussions, reviews or shared tasks from your internship.]

Skills Learned

- Using reference planes and multiple sketches to build up a more complex part.

- Applying the Flex, Shell and Fillet features and understanding their order in the feature tree.
- Checking a model in standard views and relating them to drawing views.

Outcomes

Two parts of moderate complexity were completed. The work improved confidence in planning a modelling sequence, handling many features in one part and inspecting the design from multiple viewpoints.

Week 3: Assembly Design and Project

Objectives

The objective of Week 3 was to bring individually modelled parts together into a working assembly. The assembly is also the selected internship project,

described below.

Learning Topics

- The assembly environment: inserting parts as components and organising them in the design tree.
- Fixed (f) and floating (-) components, assembly mates (such as coincident and concentric) and appearance colours.

Project Introduction

Selected project: Design and Assembly of a Wheel-and-Bracket Unit using SOLIDWORKS. The project applies everything learned in the first two weeks. The individual parts (base plate, support brackets, wheel, bush and pin) were modelled and then assembled into a single unit, shown in Figure 7.

Project Objectives

- To model each part of the wheel-and-bracket unit accurately using sketches, extrudes, cuts, fillets and ribs.
- To assemble the parts using mates so that the wheel can rotate on its axle while the base and brackets stay fixed.
- To apply colours so that every component of the assembly is easy to identify.

Scope

The project covers 3D modelling and virtual assembly of the unit. Geometry, fit and the movement of the wheel were studied in the CAD environment. The scope does not include physical manufacturing, stress analysis or load testing; these are listed as future improvements in Week 4. All dimensions are in millimetres (MMGS). [Add the overall dimensions of your assembly if required.]

Applications

A wheel mounted in a bracket on a base plate is the basic arrangement used in caster-type units, trolleys, material-handling carts, workshop equipment and rolling furniture. The same assembly method applies to pulleys, rollers and idler units in machinery.

Table 3: Main components of the wheel-and-bracket assembly

Component	Colour in model	Role in the assembly

Base plate	Blue	Foundation of the unit; carries the brackets and holds the assembly in place.
Support brackets	Green	Rise from the base with ribs and mounting holes; support the axle.
Wheel	Red	Rotating member mounted between the brackets.
Bush / spacer	Light grey	Sits between the wheel hub and the bracket to keep the wheel positioned.
Pin / axle	Gold	Passes through the bracket and hub to hold the wheel.

Assignments and Screenshots

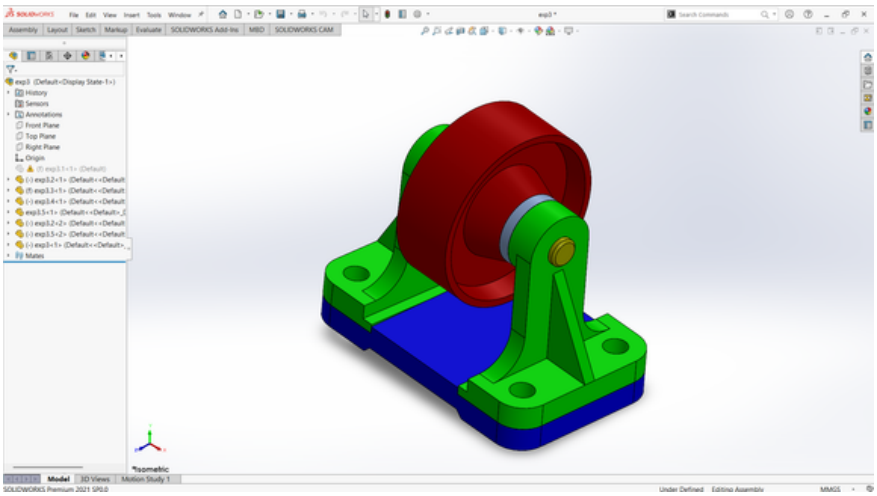


Figure 7: Wheel-and-bracket assembly (exp3) in Isometric view

Testing and Results

The assembly was checked for the status of fixed and floating components and for mate status. The components are positioned and coloured correctly, and the status bar reads Under Defined, as expected for a rotating wheel.

Challenges

Choosing which components to fix and which to leave free so that the wheel can rotate required careful use of mates. The component exp3.1<1> showed a warning symbol in the feature tree and had to be examined before the assembly could be positioned correctly.

Skills Learned

- Inserting components and positioning them with mates.
- Understanding fixed versus floating components, and diagnosing warning symbols in the design tree.

Outcomes

A complete wheel-and-bracket assembly was prepared from several individually modelled parts, bringing together the skills of Weeks 1 and 2.

Week 4: Review, Refinement and Documentation

Objectives

The objective of Week 4 was to review all the models, correct open issues and document the internship in a structured report.

Learning Topics

- Reviewing the feature trees of all models and fixing under-defined sketches and warnings.
- Capturing clear screenshots and organising the models as the project deliverable.

- Technical report writing according to the company template.

Results and Discussion

Table 4: Verification summary of the models

Figure	Model	Observation
1	Rectangle sketch	Dimensions 60 mm × 40 mm applied; status Under Defined (position not yet constrained).
2	Circle profile	Ø50 mm and Ø20 mm concentric circles; status Fully Defined.
3	L-shaped profile	Dimensions 80, 50 and 20 mm applied; status Under Defined.
4	L-bracket (Part6)	One solid body created; both holes cut successfully; feature tree without errors.
5	Bent bracket (Part1)	Holes, tabs and bend verified in four views; feature tree without errors.
6	Triangular base (exp2.1)	Shell and fillets applied; slots and body verified in four views.
7	Assembly (exp3)	Components coloured and positioned; status Under Defined; warning on exp3.1<1> to be checked.

Output

The output of the internship is a set of seven screenshots documenting sketches, solid parts and one assembly. Progress from simple to complex is visible: a plain rectangle and an L-bracket in Week 1, a bent bracket and a shelled, filleted base in Week 2, and a multi-component assembly in Week 3. Table 5 summarises the models produced.

Table 5: Models produced during the internship

Week	Model	Key features	Type
1	Part1, Part4, Part5	Rectangle, concentric circles, L-profile sketches	Sketch
1	Part6	Boss-Extrude2, 3; Cut-Extrude2, 4	Part
2	Part1	Boss-Extrude2, 3, 4; Cut-Extrude6, 7, 8; Flex3; Plane3, 5	Part
2	exp2.1	Boss-Extrude2, 3, 4; Shell1; Fillet1, 2	Part
3	exp3	Eight component entries with a Mates folder	Assembly

Future Improvements

- Fully define every sketch so that models cannot change by accident.
- Assign real materials (for example steel for the brackets and rubber for the wheel) to obtain mass properties.
- Create 2D engineering drawings with dimensions, tolerances and a bill of

materials.
- Run a motion study of the wheel, check the strength of the brackets by simulation and add fasteners for the mounting holes.

Skills Learned

Technical Skills

- 2D sketching with dimensions and geometric relations.
- 3D part modelling using extruded boss/base, extruded cut, flex, shell and fillet features.

- Use of reference planes, multi-view inspection and the feature tree.
- Assembly design with fixed and floating components, mates and appearances.

Soft Skills

The internship improved time management, because weekly assignments had to be finished within fixed periods. It also built patience and attention to detail, as small dimension mistakes in a sketch affect the whole model.

Communication

Regular discussions with the mentor required explaining modelling choices clearly and asking precise questions. Writing this report improved technical writing, including describing figures, preparing tables and citing sources. [Add an example of a presentation or review you gave, if any.]

Leadership

Taking ownership of each weekly assignment, organising the files, and helping fellow interns with sketching or assembly problems built a sense of responsibility and initiative. [Add a specific example if you led a task or helped a team member.]

Industrial Exposure

Working in a company environment showed how CAD is used in the product development cycle, where parts are modelled individually, assembled virtually and checked before manufacturing. It also demonstrated the importance of following standards, naming conventions and deadlines, and of documenting work so that others can understand and reuse it.

Outcomes

The models from Weeks 1 to 3 were consolidated, the selected project was finalised and this report was prepared. [Edit this paragraph to match the exact work you did in Week 4.]

Conclusion

Summary

The four-week internship at Nexonix Reflexo Pvt. Ltd. gave a structured introduction to 3D CAD with SOLIDWORKS Premium 2021. Week 1 covered sketching and the first solid bracket, Week 2 extended this to bent, shelled and filleted parts viewed from several directions, and Week 3 combined parts into a wheel-and-bracket assembly. Week 4 was devoted to review and documentation. Each week built on the previous one, and the resulting models show a clear growth in skill and complexity.

The experience showed that good CAD work depends on planning the modelling sequence, constraining sketches fully, keeping the feature tree clean and verifying the model in multiple views. It also showed that even a small assembly raises practical questions about constraints and movement that are not obvious from single parts.

Learning Outcomes

- Able to create and fully constrain 2D sketches using dimensions and relations.
- Able to build solid parts using extrude, cut, flex, shell and fillet features.
- Able to inspect a model in standard views and relate it to engineering drawing practice.
- Able to assemble components using mates and distinguish fixed from floating parts.
- Able to identify and fix common modelling problems such as under-defined sketches and warnings in the feature tree.
- Able to document technical work in a structured report according to a standard template.

References

References are given in IEEE format and the same sources are shown in APA format for comparison. Replace or extend these with the sources you actually used.

IEEE Format

[1] D. C. Planchard, Engineering Design with SOLIDWORKS 2021. Mission, KS, USA: SDC Publications, 2021.

[2] Dassault Systèmes, "SOLIDWORKS Help," SOLIDWORKS Documentation. [Online]. Available: https://help.solidworks.com. [Accessed: Sep. 2026].

[3] [Author initials. Surname], "[Title of paper]," [Journal name], vol. [X], no. [Y], pp. [xx–yy], [Year].

APA Format

Planchard, D. C. (2021). Engineering design with SOLIDWORKS 2021. SDC Publications.

Dassault Systèmes. (n.d.). SOLIDWORKS help. Retrieved September 2026, from https://help.solidworks.com

[Surname, Initial.] ([Year]). [Title of paper]. [Journal Name], [Volume]([Issue]), [pages].

Books

- D. C. Planchard, Engineering Design with SOLIDWORKS 2021, SDC Publications, 2021.

Websites

- SOLIDWORKS Help: https://help.solidworks.com
- SOLIDWORKS official website: https://www.solidworks.com

Research Papers

[List here only the research papers you actually read, written in the IEEE format shown above. If none were used, you may state "No research papers were referred to."]